

CSCI 4061 • LAB 02

Signals & I/O Redirection

`fork()`, signals, file descriptors, and `dup2()`



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Today's plan

- 01 Signals — what they are, and how your shell uses them
- 02 Job control — Ctrl-C, Ctrl-Z, jobs, fg, kill
- 03 File descriptors — what `fork()` and `open()` actually give you
- 04 `dup2()` — redirecting stdout, step by step
- 05 Putting it together — walking through `redirect_child.c`

What is a signal?

A short, asynchronous notification the OS delivers to a process saying that "something happened, react to it." It can arrive at any point in the program, interrupting whatever was running.

Signal	Triggered by	Default effect
SIGINT	Ctrl-C	Terminate the process (can be brought back if handler is installed)
SIGTSTP	Ctrl-Z	Pauses the process
SIGKILL	kill -9 <pid>	Terminate but cannot be caught or ignored
SIGCONT	fg / bg	Resume a stopped process

Try it: print_nums

```
while (1) {  
    printf("%d\n", i++);  
    sleep(1);  
}
```

Ctrl-C

sends SIGINT = the process
terminates immediately

Ctrl-Z

sends SIGTSTP = the process pauses,
shell says "Stopped"

fg %1

resumes counting right where it left off

Job control

Command	What it does
jobs	Lists stopped or backgrounded programs in this shell session
fg %1	Brings job 1 to the foreground, resuming it
bg %1	Resumes job 1, but keeps it running in the background
kill %1	Sends SIGTERM to job 1 so by default, terminates it

The "%1" refers to a job number.

PART 2

File Descriptors & dup2()

How a process talks to files

Every process starts with a file descriptor table

File Descriptor Table		Open Files	
0 (STDIN_FILENO)		Terminal (stdin)	
1 (STDOUT_FILENO)		Terminal (stdout)	
2 (STDERR_FILENO)		Terminal (stderr)	

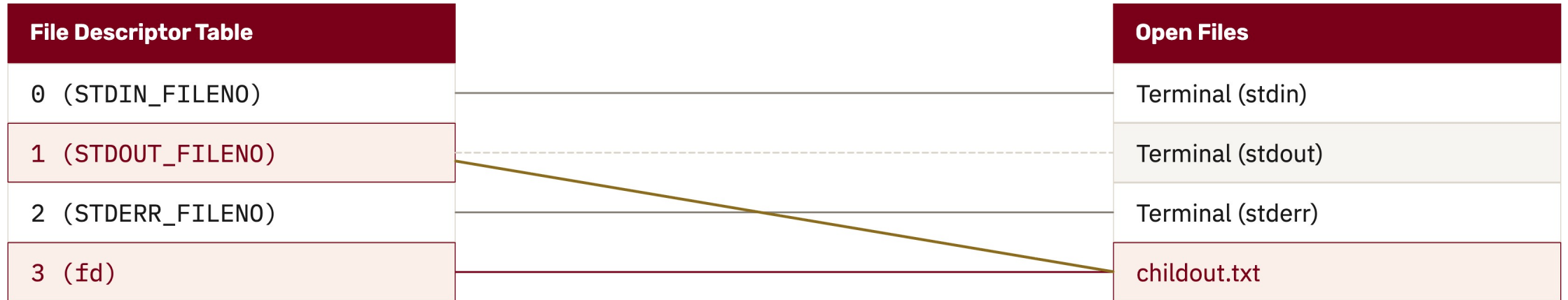
Each entry just points somewhere — right now, all three point at the terminal.

open() adds a new entry

File Descriptor Table	Open Files
0 (STDIN_FILENO)	Terminal (stdin)
1 (STDOUT_FILENO)	Terminal (stdout)
2 (STDERR_FILENO)	Terminal (stderr)
3 (fd)	childout.txt (new)

fd = open(output_file, ...) hands back the next free descriptor — 3 — pointing at the new file.

dup2(fd, 1) repoints an existing entry



fd 1 (stdout) now points at `childout.txt` too — `printf()` and the exec'd program both write straight into the file.

wc — word count

Prints the newline, word, and byte counts of a text file

Give it multiple files and it also prints a totals line

```
wc [FILE]
```

```
<newlines> <words> <bytes> <FILE>
```

wc, on an example file

```
$ cat crabapple.txt
Today I went to the crabapple tree.
I splattered 3 red and 4 green crabapples.
Life is awesome!
$ wc crabapple.txt
```

3	18	96	crabapple.txt
newlines			
	words		
		bytes	

Any questions?